



FALL PROTECTION PROGRAM

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TriCounty Metropolitan Transportation District of Oregon (TriMet)

Safety & Security Division

1020 NE 1st Ave, Portland, OR 97232

1. Policy

This program has been developed to developed to reduce, manage, and eliminate fall risks and prevent injuries related to falls.

This program is part of TriMet's environmental, health, and safety plan and serves to meet Federal OSHA and Oregon OSHA Incident Prevention Plan (IPP) and Accident Prevention Plan (APP) requirements.

TriMet's policy is to provide employees with a safe and healthy work environment, and compliance with the procedures and practices described in this program is mandatory to achieve this goal.

2. Authority

This document is administered by the Safety Department in compliance with TriMet's and includes, but is not limited to, the following regulatory rules:

- OAR Chapter 437, Division 3 Subdivision M
- 1910 Subpart I - Personal Protective Equipment
- 1910 Subpart D - Walking-Working Surfaces
- 1926 Subpart M - Fall Protection

3. Scope

This program identifies responsibilities, hazard control measures, training, inspection, and recordkeeping related to fall protection. It applies to all TriMet employees and contractors whose work functions require them to work at unprotected heights greater than 4' for General Industries and greater than 6' for the Construction Industry.

4. Positions and Responsibilities

POSITIONS	RESPONSIBILITIES
Senior Management	• Actively support this program by ensuring its contents are followed and upheld by all employees. • Provide leadership, resources, and program support. • Provide funding commensurate with the program's requirements. • Ensure individual accountability for compliance with this program. • Ensure an environment where employees are encouraged to follow these policies and procedures.
Safety Department	• Provide subject matter expertise, program oversight, and consultation to TriMet employees. • Approve manufacturers and models of Fall Protection equipment. • Determine employee training requirements and assist with training development. • Review this program at least annually to ensure it aligns with regulatory requirements and TriMet procedural changes and update as necessary.

	Periodically audit the effectiveness of the Fall Protection Program (in coordination with department managers and Supervisors who perform periodic inspections of work areas and tasks).
Managers and Supervisors	- Set the example and consistently ensure full implementation of the TriMet Fall Protection Program for TriMet employees, contractors, vendors, and visitors. - Ensure that a Job Hazard Analysis is conducted to determine the appropriate fall protection equipment for tasks assigned under their supervisor. - Provide employees with required appropriate fall protection equipment and ensure it properly fits each employee that it is assigned to. - Perform periodic inspections of work areas and tasks under their supervision to identify areas and tasks in which appropriate fall protection equipment is required and being used properly. - Ensure systems are in place to verify visitors and contractors comply with the fall protection equipment requirements found in this program. - Ensure fall protection equipment that is non-compliant, damaged, or that has been altered from the manufacturer's original specifications is removed from service. - In the case of an incident or a near-miss, inspect each affected employee's fall protection equipment for damage, and replace it when non-compliant, damaged, or if it has been altered from the manufacturer's original specification. - Ensure employees are properly inspecting and storing their fall protection equipment. - Each department shall assign a Competent Person responsible for departmental fall protection equipment program administration and maintenance including equipment inspections, inventory, and recordkeeping.
Procurement & Supply Chain Management	- The Procurement and Supply Chain Management department is responsible for sourcing, purchasing, and making available safety goods and services that are essential to TriMet's operations. - All Personal Protective Equipment procurements will be reviewed by the Safety Department to ensure the equipment meets all required Safety standards.
Engineering and Construction	Ensure all new construction and renovations, where applicable, include an engineered fall protection component as required by OAR, OSHA and TriMet Program Standards.
Competent Person	- Be certified as a Competent Person through training. - Identifies existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to employees, and who has authorization to take prompt corrective measures to eliminate them. - Be knowledgeable of OAR 437 standards and regulations related to this program. - Complete a Job Brief to identify fall hazards before Authorized Users are exposed to fall hazards. - Have the authority to stop work immediately if it is deemed unsafe to proceed. - Prepare, update, and review written fall protection plans as required. - Develop a written rescue plan when fall arrest equipment is being used. - Supervise the selection, installation, use and inspection of all non-engineered anchor points. - Verify the fall protection systems are installed and inspected in compliance with this program and applicable standards.

	• Verify all Authorized Users working at heights are trained on the specific hazards and fall protection equipment in use. • Participate in investigations of all incidents related to falls from elevated work surfaces. • Immediately remove from service any fall protection equipment found defective or subjected to forces as a result of a fall from elevated work. • Inspect fall protection equipment as recommended by the manufacturer and specified in this plan.
Authorized Users	• Read and understand the contents of this program. • Comply with the policies and procedures outlined in this document. • Use only fall protection equipment that the user has been trained on. • Comply with any further safety requirements set by supervisors or management. • Report any concerns related to fall protection to a supervisor and/or Competent Person.
Contractors	• Comply with all requirements of the Construction Safety Program (CSP), or Construction Safety Guide (CSG), included in the contract bid documents.

5. Procedures

5.1 Fall Hazard Identification

Competent Persons must determine if the walking-working surfaces on which its employees are to work have the strength and structural integrity to safely support employees. Employees may work on those surfaces only when the surfaces have the requisite strength and structural integrity.

Employees on a walking-working surface with an unprotected side or edge 4' or more above a lower level shall be protected from falling using a guardrail system, personal fall restraint, or personal fall arrest system.

If one of these systems is not available or infeasible during leading edge work, a specialized SAF-8171-PLAN-FP, Fall Protection Work Plan, must be developed and implemented to protect workers from fall hazards.

Hoist areas shall be protected by guardrails or personal fall protection systems. If guardrail systems or portions of guardrail systems are removed to facilitate the hoisting process, creating a potential fall hazard for the employee, that employee must be protected by a personal fall arrest system or personal fall restraint system.

Loading docks more than 4' above a lower level are not required to have a guardrail system on the working side of the dock where it can be demonstrated that the presence of guardrails would prevent the performance of work. A guardrail system must protect all non-working sides of a loading dock. Dock doors shall remain closed when not in use to minimize the fall hazard, and all personnel working around loading docks shall be trained to recognize and avoid the applicable fall hazards.

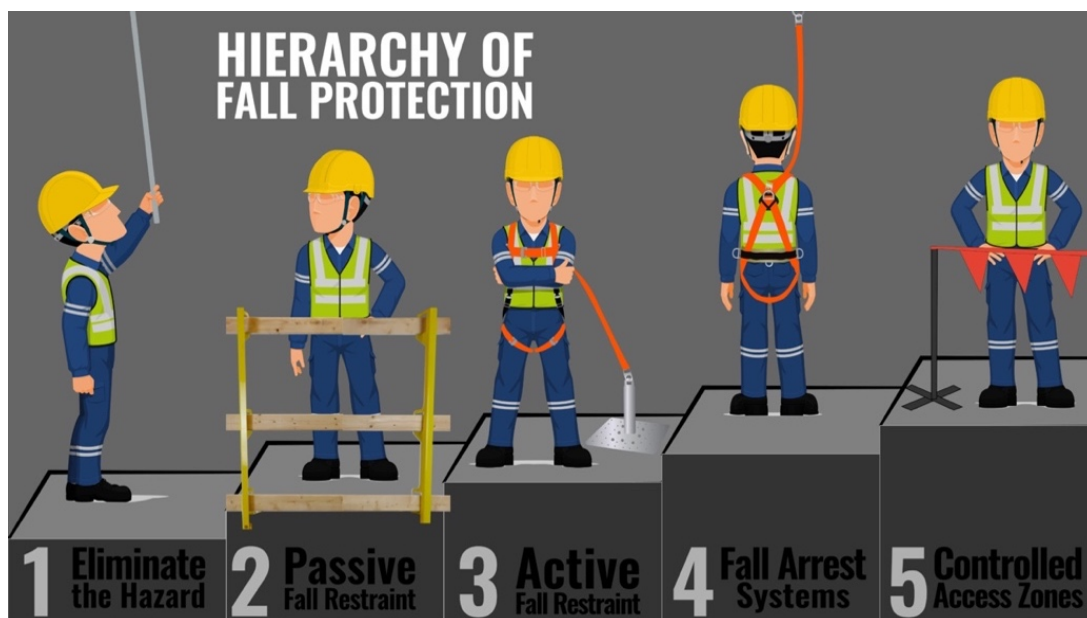
5.2 Fall Protection Selection

At a minimum, all fall protection equipment and systems used by TriMet employees and contractors must meet the OAR 437-003-1501, OSHA 1910.132 and 1926.500 requirements.

The hierarchy of Fall Protection shall be considered by the Competent Person selecting the fall protection system for a specific work practice.

All new fall protection purchased by procurement after July 1, 2025, shall be approved by the Safety Department and listed on SAF-8102-C-PPESelect, the PPE Selection Chart.

Figure 1 – Hierarchy of Fall Protection in order of preference, left to right.



Conventional fall protection systems provide the greatest protection against fall hazards and should be considered first. These include:

- Standard guardrail system
- Fall restraint system
- Personal fall arrest system
- Warning line system

5.3 Holes and Skylights

Regardless of height, each employee on a walking-working surface must be protected from tripping in or stepping into or through holes (floor openings, holes, manholes, roof hatches, and skylights) by covers or equivalent, or by personal fall protection systems (OAR 437-003-1501).

Hole or Skylight covers:

- Must be secured to prevent accidental displacement.

- Must be able to withstand a load of at least twice the maximum intended load.
- All covers shall be color-coded or marked with the words “HOLE” or “COVER.”
- Covers located in roadways and vehicular aisles shall be capable of supporting, without failure, at least twice the maximum axle load of the largest vehicle expected to cross over the cover.

Where covers are utilized as fall protection measures, they shall remain in position when not in use. If covers are removed, the fall hazard must be monitored by a designated employee who is safeguarded by a fall protection system or travel restraint system.

Where skylights are installed, their design must comply with applicable building codes and should be documented to ensure sufficient strength to prevent a fall, installed with a skylight cover or be secured with a guardrail system.

5.4 Dangerous Equipment

Employees working above hazardous equipment, shall be safeguarded from potential falls into or onto the equipment by means of protective equipment guards, guardrail systems, warning line systems or personal fall restraint systems.

5.5 Rolling Stock

All employees must be protected from fall hazards when working on motor and rolling stock vehicle surfaces more than 10 feet above a lower level or at any height above dangerous equipment.

5.6 Excavations

Employees engaged in activities at the edge of excavations that exceed 6' in depth must be safeguarded against potential falls through the implementation of guardrail systems, fences, barricades, or personal fall arrest systems. This requirement encompasses wells, pits, shafts, and other similar excavations.

Excavations of any height, where the public may be exposed, shall be safeguarded against fall hazards through the implementation of guardrail systems, fences or barricades.

5.7 Building Rooftops

All new construction, renovation or repair to existing roof systems shall incorporate engineered fall protection into the project design. This includes guardrail systems (including parapets that meet the height requirements) and/or anchors for personal fall arrest or fall restraint systems. The design of fall protection shall be approved by a Professional Engineer.

Only Competent Persons, or Authorized Users under the supervision of a Competent Person, may perform work on buildings with fall restraint or fall arrest protection installed.

Where a building lacks a permanent guardrail system (including parapets) or a fall arrest/restraint anchor for rooftop fall protection, before employees access the rooftop, a

Competent Person must develop a Fall Protection Work Plan based on the tasks being performed.

5.8 Guardrail Systems and Parapets

Guardrail systems both provide a barrier to prevent employees from falling to lower levels and designate an area where work may take place without the use of additional fall protection PPE.

Guardrails must be installed along each unprotected side or edge.

Where guardrail systems are in place as a fall protection measure, the railing must have a vertical height of 42" plus or minus 3" measured from the upper surface of the top rail to the working surface and must consist of a top rail, an intermediate rail, and posts. Mid-rails, screens, mesh, intermediate vertical members, or equivalent intermediate structural members must be installed between the top edge of the guardrail system and the walking or working surface when there is no wall or parapet wall at least 21" high.

Guardrails must be designed to prevent injury to an employee from punctures or lacerations and to avoid snagging of clothing.

The intermediate rail should be placed about halfway between the top rail and the working surface.

Guardrail systems must withstand a force of at least 200 lbs. in an outward or downward without failure. When a force of 200 lbs. is applied downward, the top edge of the guardrail must not deflect to a height lower than 39" above the working surface. The ends of all top rails and mid rails cannot overhang the terminal posts, except where any overhang does not pose a strike, impalement, or tripping hazard.

Top rails and mid rails must have a diameter or thickness of at least 0.25" to prevent cuts and lacerations. If wire rope is used for top rails, it should be flagged at intervals no greater than 6' with high-visibility material.

Intermediate vertical members must be installed no more than 19" apart.

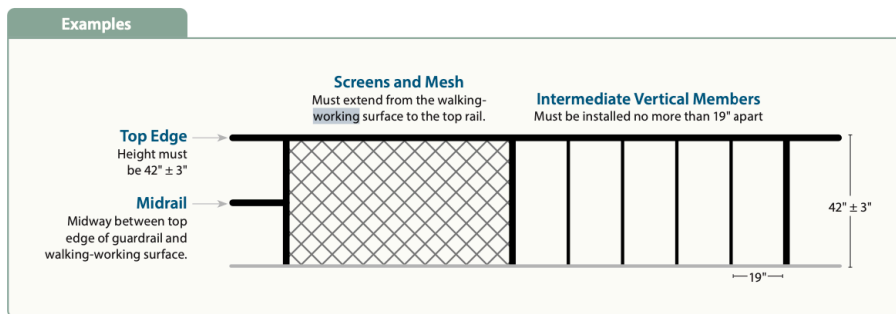
A standard toeboard shall be installed on all guardrail systems where people can pass underneath the elevated work surface, where there is moving machinery, and/or where equipment is used on the elevated surface that poses a risk of falling equipment. Toeboards must be a minimum of 3.5" in vertical height and must be securely fastened with no more than 0.25" of clearance above the working surface.

If materials are stored near the guardrail system at heights exceeding the toeboard, paneling must be provided from the work surface to the intermediate rail.

Toeboards shall be capable of withstanding, without failure, a force of at least 50 lbs. applied in any downward or outward direction at any point along the toeboard.

Engineered guardrail systems may be used if they meet these requirements and are installed according to the manufacturer's specifications.

Portable guardrail systems may be utilized as a fall protection measure, provided they meet the requirements of OAR, OSHA, and ANSI/ASSP guardrail specifications.



Parapets are considered equivalent guardrail system members, if they are at least 39" above the walking-working surface. The top edge height may exceed 45" provided the guardrail system meets all other criteria of this section.

5.9 Personal Fall Restraint Systems

Personal fall restraint systems shall be rigged to prevent the user from falling any distance.

TriMet requires a full body harness to be used in a fall restraint system. The attachment point to the full body harness may be at the back, front or utilizing both side D-rings.

All components of the fall restraint system, including connectors, D-rings, snap hooks, lanyards, and body harnesses, must comply with all applicable ANSI/ASSP and OSHA requirements and must have a minimum breaking strength of 5000 lbs.

The gate strength of carabiners and snap hooks must be able to withstand a minimum load of 3600 lbs. (16 kN). Snap hooks and carabiners must be the automatic locking type that require at least 2 separate, consecutive movements to open.

Snap hooks and carabiners must not be connected to any of the following unless they are designed for such connections:

- Directly to webbing, rope, or wire rope
- To each other
- To a D-ring to which another snap hook, carabiner, or connector is attached
- To a horizontal lifeline
- To any object that is incompatibly shaped or dimensioned in relation to the snap hook or carabiner such that unintentional disengagement could occur when the connected object depresses the snap hook or carabiner gate, allowing the components to separate.

Anchorage used for attachment of personal fall restraint equipment shall be independent of any anchorage being used to support or suspend platforms and shall be capable of supporting 3000 lbs. (13.3 kN) per employee attached, or be designed, installed and used as follows:

- As part of a complete personal fall restraint system which maintains a safety factor of at least two, and
- Under the supervision of a Qualified Person.

The restraint lanyard must be short enough to prevent a fall from occurring, be protected against cutting and abrasion.

5.10 Positioning Systems

When used as a positioning device system, the lanyard length shall be rigged such that an employee cannot free fall more than two feet.

Positioning devices must be secured to an anchorage that can support at least twice the potential impact load of an employee's fall, or a minimum of 3000 lbs.

All components of the positioning system, including connectors, D-rings, snap hooks, lanyards, and body harnesses, must comply with all applicable ANSI/ASSP and OSHA requirements and must have a minimum breaking strength of 5000 lbs.

The gate strength of carabiners and snap hooks must be able to withstand a minimum load of 3600 lbs. (16 kN). Snap hooks and carabiners must be the automatic locking type that require at least 2 separate, consecutive movements to open.

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- To each other
- To a D-ring to which another snap hook, carabiner, or connector is attached
- To a horizontal lifeline
- To any object that is incompatibly shaped or dimensioned in relation to the snap hook or carabiner such that unintentional disengagement could occur when the connected object depresses the snap hook or carabiner gate, allowing the components to separate.

5.11 Personal Fall Arrest System

These systems are used to prevent injuries to employees in case a fall from an elevated work surface occurs. The use of a fall arrest system requires employees to wear a full body harness and use a deceleration device, also called a shock pack, to reduce forces on the body during a fall. Body belts are prohibited with fall arrest systems.

All components of the fall arrest system, including connectors, D-rings, snap hooks, lanyards, and body harnesses/belts, must comply with all applicable ANSI/ASSP and OSHA requirements and must have a minimum breaking strength of 5000 lbs.

The gate strength of carabiners and snap hooks must be able to withstand a minimum load of 3600 lbs. (16 kN). Snap hooks and carabiners must be the automatic locking type that require at least two separate, consecutive movements to open.

Snap hooks and carabiners must not be connected to any of the following unless they are designed for such connections:

- Directly to webbing, rope, or wire rope
- To each other
- To a D-ring to which another snap hook, carabiner, or connector is attached
- To a horizontal lifeline
- To any object that is incompatibly shaped or dimensioned in relation to the snap hook or carabiner such that unintentional disengagement could occur when the connected object depresses the snap hook or carabiner gate, allowing the components to separate.

Anchorage used for attachment of personal fall arrest equipment shall be independent of any anchorage being used to support or suspend platforms and shall be capable of supporting 5000 lbs. (13.3 kN) per employee attached, or be designed, installed and used as follows:

- As part of a complete personal fall restraint system which maintains a safety factor of at least two, and
- Under the supervision of a Qualified Person.

Employees utilizing personal fall arrest systems shall not perform work alone.

Lifeline systems used as a component of a fall arrest system shall be designed and installed under the supervision of a qualified person and used under the supervision of a Competent Person.

Lifelines or other components of a fall arrest system must not be attached to guardrail systems, ladders, scaffolding components, building fixtures, conduits or plumbing, other lanyards, roof stacks, vents, pipes, or any unauthorized anchor points.

Lifelines shall be protected from cuts and abrasions.

When stopping a fall, personal fall arrest systems shall:

- Be attached to an anchor point capable of supporting at least 5000 lbs. per employee attached.
- Limit maximum arresting force on an employee to 1800 lbs.

- Ensure employees cannot free fall more than 6' or make contact with any lower level due to a fall.
- Bring an employee to a complete stop and limit the maximum deceleration distance to 3.5'.
- Be able to withstand twice the potential impact energy of an employee falling from a height of 6' or the maximum fall distance allowed by the system, whichever is less.

Personal fall arrest systems must be worn with the attachment point of the body harness located in the center of the employee's back near shoulder level. The attachment point may be located in the pre-sternal position if the free fall distance is limited to 2' or less.

If the Personal Fall Arrest system is used by an employee having a combined body and tool weight of 310 lbs. or more and the Safety Department must be contacted to select a Fall Protection system that would be most appropriate.

Fall protection equipment may not be used to hoist equipment or tools to an elevated work surface.

Personal fall arrest systems and components that have been subjected to impact loading must be immediately removed from service.

5.12 Warning Line Systems

Warning line systems typically consist of a rope, wire, or chain attached to a stanchion to delineate a designated work area on a low sloped roof. The warning line will be marked every 6' with high-visibility material.

Only commercial produced warning line systems may be used by TriMet employees.

A warning line system shall not be used as fall protection on roof slopes greater than 2/12 (vertical to horizontal).

Warning line systems shall be erected around all open sides of the roof work area is 6' from the roof edge when roofing work is being performed.

If mechanical equipment is in use on the rooftop, the warning line must be at least 6' from the roof edge in line with the direction of equipment operation, and at least 10' from the roof edge perpendicular to the equipment's direction of operation.

When work other than roofing work is taking place, the warning line must be erected at least 15' from the unprotected sides or edges.

Access points, material handling zones, storage spaces, and hoisting areas must be clearly marked and linked to the work area by an access path created with two warning lines.

When the path or point of access is not in use, a rope, wire, chain or other barricade equivalent in strength and height to the warning line shall be placed across the path.

Warning lines should consist of ropes, wires, or chains and be supported by stanchions.

The line shall be supported to ensure the lowest point is not less than 34" above the work surface and not more than 39" at its highest point.

After being erected, with the rope, wire, or chain attached, stanchions shall be capable of resisting, without tipping over, a force of at least 16 lbs. applied horizontally against the stanchion, 30" above the walking-working surface, perpendicular to the warning line, and in the direction of the floor, roof, or platform edge.

The rope, wire, or chain must have a minimum tensile strength of 500 pounds. After being attached to the stanchions, it should be able to support the loads applied to the stanchions without breaking.

The line shall be attached at each stanchion in such a way that pulling on one section of the line will not result in slack being taken up in adjacent sections.

Employees are not permitted to enter the area between the roof edge and warning line unless work is being conducted on that portion of the roof and adequate fall protection measures are in place, such as a fall restraint system, fall arrest system or a Safety Monitor.

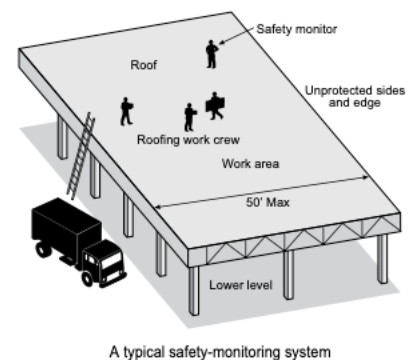
5.13 Safety Monitoring System OAR 437-003-2502

Safety monitoring systems for roofing work and their use shall comply with the following provisions.

- A safety monitoring system shall not be used as a fall protection system for any work other than roofing work on roof slopes of 2/12 (vertical to horizontal) or less.
- The use of a safety monitoring system alone (i.e. without the warning line system) is not permitted on roofs more than 50' in width.

The employer shall designate a Competent Person to monitor the safety of other employees, and the employer shall ensure that the safety monitor complies with the following OAR 437-003-2502 requirements:

- The safety monitor shall be competent to recognize fall hazards.
- The safety monitor shall warn the employee when it appears that the employee is unaware of a fall hazard or is acting in an unsafe manner.
- The safety monitor shall be on the same walking-working surface and within visual sighting distance of the employee being monitored.
- The safety monitor shall be close enough to communicate orally with the employee.
- The safety monitor shall not have other responsibilities which could take the monitor's attention from the monitoring function.



Mechanical equipment shall not be used or stored in areas where safety monitoring systems are being used to monitor employees engaged in roofing operations.

No employee, other than an employee engaged in roofing work, shall be allowed in an area where an employee is being protected by a safety monitoring system.

5.14 Mobile/Temporary Anchor Point

Temporary anchor points may be used in certain areas or structures where work needs to be performed within 10' of a roof edge and no fall protection equipment is available on the roof.

A Competent Person must oversee the setup and use of temporary anchor points.

Temporary anchor points must comply with all applicable OAR standards for fall protection anchor points.

Lanyards utilized with a temporary anchor point must not introduce additional hazards to the worker.

5.15 Protection from Falling Objects

When elevated work is taking place and there is a risk of falling objects posing a hazard to individuals on lower levels of an elevated work surface, precautions must be taken to prevent injuries from occurring.

At TriMet facilities, where employees may be exposed to hazards associated with falling objects, one of the following methods of protection for falling objects must be implemented.

- Helmets are worn by employees who may pass through or work in an area where employees may be exposed to hazards associated with falling objects.
- Toeboards, screens or guardrail systems are placed on the elevated work surface to prevent objects from falling.
- A canopy structure, capable of withstanding a falling object without collapse or penetration, is erected to keep potential falling objects far enough from the edge or the elevated work surface to avoid creating a fall hazard.
- The area where objects can fall is barricaded, and access to it is not authorized when work is being done on an elevated work surface.
- Where tools, equipment, or materials are stacked higher than the top edge of a toeboard, paneling or screening must be erected from the walking-working surface or toeboard to the top of the guardrail system's top rail or mid rail, extending far enough to protect individuals below.

When in construction work, including demolition, or power transmission distribution and maintenance work is being done, where employees may be exposed to hazards associated with falling objects, helmets must be worn and one of the preceding methods of protection for falling objects must be implemented.

When an elevated work area poses a potential fall hazard where the public might be exposed to falling object risks, one of the following precautions should be implemented.

- Redirect public traffic through a barrier system to ensure they do not enter areas where falling object hazards exist.
- Erect a structure capable of withstanding impact from a fallen object under which the public may travel.

5.16 Rescue Operations

TriMet is responsible for ensuring prompt rescue of employees in the event of a fall or assuring that employees can rescue themselves.

Self-rescue: If the fallen employee can rescue themselves, they can utilize existing fall protection equipment or self-provided rescue equipment.

Assisted Rescue: If the fallen employee is unable to perform self-rescue, other trained personnel ensure the fallen employee is brought to safety using adequate means.

When a personal fall arrest system is utilized as a fall protection measure, the Competent Person must use a Fall Protection Work Plan to document rescue plans to ensure employees can be safely rescued after a fall in a timely manner.

In a fire department urban response area, calling 911 immediately after a fall can be utilized as a rescue plan.

In remote areas, or where the local fire department's response time is estimated to be more than 10 minutes, an employee administered rescue plan must be created by a Competent Person.

5.17 Exemptions

Employees are exempt from attaching to a fall protection system only under the following temporary and infrequent conditions:

- During initial installation of the fall protection anchor (prior to engaging in any work activity), or the disassembly of the fall protection anchor after the work has been completed.
- An employee directly involved with inspecting, assessing, or estimating roof-level conditions only on low pitched roofs prior to the actual start of construction work or after all construction work has been completed.

Examples of activities recognized as inspecting or estimating include:

- Measuring a roof to determine the amount of materials needed for a project.
- Inspecting the roof for damage without removing equipment or components.
- Assessing the roof to determine what method of fall protection will be provided for employees.

This exemption does not apply on steep pitch roofs, where construction work is underway, or when fall protection systems or equipment have been installed and are available for workers to use for pre-work and post-work inspections, investigations, or assessments.

TriMet requires the use of a Safety Monitor when using a fall protection system exemption.

If using a conventional fall protection exemption, a Competent Person must complete a SAF-8171-PLAN-FPWORK, Fall Protection Work Plan, explaining why it was infeasible to use conventional fall protection, and document the use of a Safety Monitor during this temporary and infrequent work.

6. Training

TriMet shall provide a training program for each employee who might be exposed to fall hazards.

All trained employees must have a written certification record that meets the OAR 437-003-05503. The latest training certification shall be maintained.

Employees must be trained to the Competent Person or the Authorized User level. Authorized Users must operate under the supervision of a Competent Person.

Retraining is required every 3 years, or when:

- Changes in the workplace render the previous training obsolete, or
- Changes in the types of fall protection systems or equipment render the previous training obsolete, or
- Inadequacies in an affected employee's knowledge or use of fall protection systems or equipment indicate that the employee has not retained the requisite understanding or skill.

Training will be assigned to each group as follows:

POSITIONS	GUIDE NUMBER	TRAINING TOPIC	FREQUENCY
Employees who select a form of Fall Protection	SAF-8174-TG-FPCP	Fall Protection Competent Person	Every 3 years
Employees who are authorized for certain types of Fall Protection	SAF-8175-TG-FPAU	Fall Protection Authorized User	Every 3 years

7. Equipment and inspection requirements

7.1 Protection Equipment Selection

The SMS Manager or designee is responsible for selecting all personal fall protection systems and equipment. A list of approved Fall Protection Equipment is found in SAF-8102-C-PPESelect.

All personal fall protection equipment selected and purchased after July 1, 2025, shall be meet the Standards in Table 11 – Fall Protection Equipment.

7.2 Personal and Temporary Fall Protection Equipment Inspection and Maintenance

Fall Protection equipment must be provided, used, and maintained per manufacturer's recommendations.

Managers and Supervisors are responsible for maintaining all fall protection systems across TriMet facilities. This is achieved by designating a Competent Person trained in relevant areas or by employing contractors experienced in these operations.

NOTE: If an employee is involved in an incident or accident involving a fall from an elevated work surface, the permanent fall protection system components must be removed from service and inspected by a Qualified Person who can provide service and certify that the system is safe for use. Harnesses, lanyards, or other PPE involved in a fall incident may not be returned to service. PPE involved in a fall or placed under tension must be rendered useless and discarded.

Personal fall protection systems must be inspected by the wearer before initial use of each work shift for mildew, wear, damage, and other deterioration, and defective components must be removed from service.

Temporary fall protection equipment, such as warning lines, must be inspected by a Competent Person upon installation and before use. If the system experiences tension due to an accident or near miss, it should be reinspected to ensure it meets all applicable requirements.

An annual fall protection equipment inspection must be completed by a Competent Person documented using an SAF-8173-F-FPEQUIP, Fall Protection Equipment Inspection checklist. Managers must retain copies of for 1 year.

Modification of commercial fall protection equipment is prohibited.

7.3 Permanently Installed Fall Protection Equipment Inspection

Fall protection permanent systems must be inspected annually by a Competent Person. Annual inspections should address all components of a fall protection system, including, but not limited to, anchor points, lifelines, structural components, guardrails, parapets, and personal protective equipment. Any deficiencies identified during the inspection or certification process must be addressed by a Qualified Person prior to an employee using the system.

An inventory of all fall protection systems on working unit buildings should be maintained. This can be completed on a SAF-8172-F-FPFAC, Fall Protection Facility Inspection form, by a site manager or their designee.

7.4 Fall Protection Equipment Storage

Fall protection equipment, including restraint lanyards and body harnesses, should be stored in a well-ventilated, clean, dry area, free from temperature and humidity extremes, corrosive materials, and other contaminants.

TABLE 10 – FALL PROTECTION EQUIPMENT SELECTION			
HAZARD CONTROL	CLASSIFICATION	PROTECTORS	INSPECTION FREQUENCY
Falls > 4' Fall Arrest	<u>ANSI/ASSP Z359.11</u> Anchors - 5000 lb. or 2:1 Safety Factor	Permanent Anchorage	- Before each use. - Annual – Documented.
Falls > 4' Fall Arrest	<u>ANSI/ASSP Z359.11</u> Temporary Anchors - 5000 lb.	Temporary or Mobile Anchorage	- Before each use. - Annual – Documented.
Falls > 4', no OCS exposure Fall Arrest/Restraint/Positioning	<u>ANSI/ASSP Z359.11</u> Body Harness - Non-Arc Flash Rated	Full Body Harness – Non Arc Flash	- Before each use. - Annual – Documented.
Falls > 4', OCS exposure within 10' Fall Arrest/Restraint/Positioning	<u>ANSI/ASSP Z359.11</u> Body Harness - Arc Flash Rated	Full Body Harness – Non Arc Flash	- Before each use. - Annual – Documented.
Falls > 4', no electrical hazards Fall Arrest/Restraint	ANSI/ASSP Z359.15 Vertical Lifeline – Non-Arc Flash Rated 5000 lb. anchor required	Vertical lifeline	- Before each use. - Annual – Documented.
Falls > 4', no electrical hazards Fall Arrest/Restraint	ANSI/ASSP Z359.15 Rope Grab and Shock Pack – Non-Arc Flash Rated 5000 lb. anchor required	Rope Grab and Deceleration Pack	- Before each use. - Annual – Documented.
Falls > 4', no OCS exposure Fall Arrest	<u>ANSI/ASSP Z359.14-2021</u> Self-Retracting Lanyard Class 1 or B – No OCS Exposure 5000 lb. anchor required Must be anchored above the dorsal D-ring.	Self-Retracting Lanyard – Single Leg Self-Retracting Lanyard – Double Leg	- Before each use. - Annual – Documented.

Falls > 4', OCS exposure within 10' Fall Arrest	<u>ANSI/ASSP Z359.14-2021</u> Self-Retracting Lanyard Class 2 – FR Rated 5000 lb. anchor required May be connected to 5' below the dorsal D-ring	Self-Retracting Lanyard – Single Leg – Arc Flash Rated Self-Retracting Lanyard – Double Leg – Arc Flash Rated	• Before each use. • Annual – Documented.
Falls > 4', OCS exposure within 10' Positioning/Fall Restraint	ANSI/ASSP Z359.3 Positioning Lanyard - Arc Flash Rated 3000 lb. anchor required May not allow more than a 2' free fall	Positioning Lanyard – Arc Flash Rated	• Before each use. • Annual – Documented.

8. Record Keeping, Forms and Program Review

8.1 Program Review

This program's procedures must be reviewed and updated at least every 3 years.

All employees exposed to fall hazards must maintain a written certification of training.

The certificate should include the employee's name, the date of training, and the signatures of trainer, the employer, and authorizing manager or supervisor.

All inspections and certification records for fall protection systems and PPE must also be maintained for 1 year by the site or department manager.

Written fall protection procedures should be developed and maintained for each operation conducted within the department to address measures taken to eliminate fall hazards.

The Competent Person must develop and maintain written rescue operations for each system or project with a fall hazard.

8.2 Forms

This Standard requires the following forms, and each are considered part of the adopted program. Forms may be updated as necessary with approval from the SMS Manager.

All program documents and forms may be accessed on TriNet under the Safety & Security division page.

FORM NUMBER	TITLE	RESPONSIBLE PARTY	DOCUMENT RETENTION
SAF-8171-PLAN-FP	Fall Protection Work Plan	Competent Person	Duration of Project
SAF-8172-F-FPFAC	Fall Protection Facility Inspection	Qualified Person Site Manager	1 Year
SAF-8173-F-FPEQUIP	Fall Protection Equipment Inspection	Competent Person	1 Year
SAF-8174-TG-FPCP	Fall Protection Competent Person Training Guide	Safety Department	3 Years
SAF-8175-TG-FPAU	Fall Protection Authorized User Training Guide	Safety Department	3 Years

9. Definitions

TERM	DEFINITION
Authorized Person	An employee using fall protection equipment who has been trained on this policy and the use of specific fall protection equipment, who works under the direction of a Competent Person regarding the use of fall protection and rescue systems.
Anchor Point/ Anchorage	A secure point of attachment for lifelines, lanyards or deceleration (grabbing) devices.
Body Harness (also referred as full-body harness)	An interconnected set of straps that is secured about a person in a manner distributing the fall arrest forces over at least the thighs, pelvis, waist, chest and shoulders, with a means for attaching the harness to other components of a personal fall arrest system.
Connector	A device that is used to connect parts of a personal fall arrest system together (i.e. D-rings and snap hooks).
Competent Person	A trained individual designated by the employer to be responsible for the immediate supervision, implementation and monitoring of the employer's managed fall protection program who, through training and knowledge, can identify, evaluate and addressing existing and potential fall hazards, and who has the employer's authority to take prompt corrective action about such hazards.
Deceleration Device	The additional vertical distance a falling person travels, excluding lifeline elongation and free fall distance, before stopping, from the point at which a deceleration device begins to operate. It is measured as the distance between the location of the individual's body harness attachment point at the moment of activation (at the onset of fall arrest forces) of the deceleration device during a fall, and the location of that attachment point after the individual comes to a full stop.
Fall Arrest	Equipment used to stop a fall after someone has fallen. It is designed to stop someone before they strike a lower level.
Fall Restraint	Equipment used to keep someone from approaching a fall hazard. The lanyard must be short enough to keep the individual from reaching the leading edge.

Fixed Ladder	A ladder, including individual rung ladders, which are permanently attached to a structure, building, or equipment. It does not include ship's stairs or manhole steps.
Guardrail	A barrier erected to prevent personnel from falling to lower levels.
Hazardous Slope	A slope from which construction work is performed where normal footing cannot be maintained without the use of devices due to the pitch of the surface, weather conditions, or surface material.
Hole	A void or gap 2" or more in its least dimension in a floor, roof, or other walking-working surface.
Horizontal Lifeline (HLL)	A cable or rail that is tensioned and secured between two anchor points. Horizontal lifelines should be designed and installed under the supervision of a Qualified Person.
Infeasible	Situations where it is impossible to perform the construction work using a conventional fall protection system (i.e. guardrail system, safety net system, or personal fall arrest system) or that it is technologically impossible to use any one of these systems to provide fall protection.
Lanyard	A flexible line of rope, wire rope, or strap, which generally has a connector at each end for connecting the body harness to a deceleration device, lifeline or anchorage.
Leading Edge	The advancing edge of a floor, roof, or formwork for a floor or other walking-working surface (such as the deck) that changes location as additional floor, roof, decking, or formwork sections are placed, formed, or constructed.
Lower Levels	Those areas or surfaces to which an employee can fall. Such areas include, but are not limited to, ground levels, floors, platforms, ramps, runways, excavations, pits, tanks, material storage, water, equipment, structures, or portions thereof.
Low-slope Roof	A roof having a slope less than or equal to 2/12 (vertical to horizontal).
Maximum Intended Load	The total load (weight and force) of all employees, equipment, vehicles, tools, materials, and other loads the employer reasonably anticipates being applied to a walking-working surface at any one time.
Mechanical Equipment	All motor or human propelled wheeled equipment used for work, excluding wheelbarrows and mop carts.
Opening	A gap or void 30" or higher than 18" or wider in a wall or partition, through which personnel can fall to a lower level.
Orthostatic Intolerance	The development of symptoms such as light-headedness, palpitations, tremulousness, poor concentration, fatigue, nausea, dizziness, headache, sweating, weakness and occasionally fainting during upright standing.
Personal Fall Arrest System (PFAS)	A system used to arrest an employee in a fall from a working level. It consists of an anchorage, connectors, and body harness and may include a lanyard, deceleration device, lifeline, or suitable combinations of these.
Positioning Device System	A body harness system rigged to allow an employee to be supplied on an elevated vertical surface, such as a wall, and work with both hands free while leaning.
Qualified Person	One with a recognized degree or professional certificate (e.g. Professional Engineer PE) or someone with extensive knowledge, training and experience in the fall protection and rescue field who is capable of designing, analyzing, evaluating and specifying fall protection and rescue systems to the extent required by these standards.

Restraint Line	A device, which is attached between the employee and an anchorage to prevent the employee from walking or falling off an elevated surface.
Roof	The exterior surface on the top of a building.
Roofing Work	The hoisting, storage, application, and removal of roofing materials and equipment, including related insulation, sheet metal, and vapor barrier work, but not including the construction of the roof deck.
Rope Grab	A deceleration device that travels on a lifeline and, by friction, engages on the lifeline to arrest a fall.
Safety Monitor System	A fall protection measure for low slope roofing work wherein a Competent Person physically monitors work and warns workers when they are close to a fall hazard.
Self-retracting Lifeline/ Lanyard	A fall arrest device containing a drum-wound line that can be slowly extracted from, or retracted onto, the drum under minimal tension during normal movement and which, after onset of a fall, automatically locks the drum and arrests the fall (usually within 2' or less).
Snap Hook	A connector consisting of a hook-shaped member with a normally closed keeper, or similar arrangement, which may be opened to permit the hook to receive an object and automatically closes when released to retain the object.
Steep Roof	A roof having a slope greater than 2/12 (vertical to horizontal).
Suspension Trauma	A description of a combination of effect which occurs when the human body is held upright without any movement for a period of time.
Toeboard	A low protective barrier at least 3.5" in height that prevents material and equipment from falling to a lower level.
Vertical Lifeline	A fall arrest system component consisting of a rope or wire connecting other components of a personal fall arrest system to the anchor point.
Walking-working Surface	Any surface, whether horizontal or vertical on which an employee walks or works, including, but not limited to, floors, roofs, ramps, bridges, runways, form work and concrete reinforcing steel but not including ladders, vehicles, or trailers, on which employees perform their job duties.
Warning Line System	A fall restraint system consisting of a barrier erected on a roof to warn employees that they are approaching an unprotected roof side or edge, and which defines a designated area in which roofing work may take place without the use of guardrail or other systems to protect employees in the area.
Work Area	That portion of a walking-working surface where job duties are being performed.
ACRONYM	TERM
ANSI	ANSI American National Standards Institute
ASTM	ASTM American Society for Testing and Materials
ASSP	ASSP American Society of Safety Professionals
CFR	CFR Code of Federal Regulations
EC&P	EC&P Engineering, Construction, and Planning (TriMet)
JHA	JHA Job Hazard Analysis
PFAS	PFAS Personal Fall Arrest System

PE	PE Professional Engineer
PPE	PPE Personal Protective Equipment
OAR	Oregon Administrative Rules
OSHA	OSHA Occupational Safety and Health Administration
SRL	SRL Self-Retracting Lifeline

10. References

REFERENCE	CITATION	COMMENTS/LINKS
OAR	437-003-Subdivision M	Fall Protection
OAR	437-003-1501 437-003-2501 437-003-0502 437-003-1502 437-003-2502 437-003-0503	Fall Protection Protection from Falling Objects Personal Fall Restraint Warning Line Systems for Roofing Work Safety Monitoring Systems Training Requirements
OSHA	1910 Subpart I 1910 Subpart D 1926 Subpart M	Personal Protective Equipment Walking-Working Surfaces Fall Protection
ANSI	ANSI Z89.-2009 ANSI Z41-1999 ANSI Z359.11 ANSI Z359.3-2019 ANSI/ASSE Z359.12-2002	Industrial Head Protection Protective Footwear Full Body Harnesses Safety Requirements for Lanyards Connecting Components For Personal Fall Arrest Systems

11. Version History

VERSION	DATE	SUMMARY OF CHANGES	OWNER	APPROVED BY
1.0	3/17/25	Original Version	SMS Manager Jessie Harvey	